MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)
FIFTH SEMESTER EXAMINATION
(2024 ADMISSION ONWARDS)
Course Code: **MG5DSEBOT300** Course Title: **Plant breeding and plant genetic resources**

Time: 2 Hrs **Maximum Marks:** 70

Taxonomy Levels: Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I), and Appreciation (Ap)

Student should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A: Very Short Answer Type Questions

Answer any 10 questions. Each question carries 2 marks

No.	Question	Taxonomy	CO
1	Who proposed the Centres of Origin of Cultivated Plants?	Remember	1
2	Expand ICAR and NBRI.	Remember	1
3	Define acclimatization in plant introduction.	Remember	2
4	Which plant breeding method would you choose to develop disease-resistant lines from a local heterogeneous landrace?	Apply	2
5	What is meant by mutation breeding?	Understand	2
6	Define heterosis.	Remember	3
7	Mention any two harmful effects of inbreeding depression.	Understand	3
8	Name any two traditional rice varieties cultivated in Kerala.	Remember	4
9	Write the binomial and common name of any one spice plant mentioned in your syllabus.	Remember	4
10	Give the common name and identify the edible part of *Artocarpus hirsutus*.	Apply	4
11	Define deforestation as a threat to plant genetic resources.	Understand	5
12	Name two *in-situ* conservation methods.	Remember	5

(10×2=20 marks)

Part B: Short Answer Type Questions

Answer any 8 questions. Each question carries 5 marks

13	Write a brief note on the significance of plant breeding in crop improvement.	Remember	1

14	Mention the location and major crop achievements of CPCRI and CTCRI in Kerala.	Understand	1
15	Briefly explain the basic steps involved in the plant introduction procedure.	Understand	2
16	Outline the procedure of artificial hybridization and explain how emasculation prevents self-pollination.	Apply	2
17	What is polyploidy breeding? Mention its applications in crop plants.	Understand	2
18	Explain the Dominance hypothesis of heterosis.	Understand	3
19	Outline the procedure of the Pedigree breeding method.	Understand	3
20	Discuss the role of traditional knowledge in conserving plant genetic resources.	Apply	4
21	List the binomial, family, and economic uses of *Averrhoa carambola* and *Sauropus androgynus*.	Remember	4
22	Briefly describe Major threats to the genetic resources.	Understand	5

(8×5=40 marks)

Part C: Essay Type Questions

Answer any 1 question. Question carries 10 marks

23	Explain the following breeding methods with their merits and demerits: (a) Mass selection, and (b) Clonal selection.	Understand	2
24	Describe the major *in-situ* and *ex-situ* methods used for the conservation of plant genetic resources.	Understand	5

(1×10=10 marks)